Behind the Lens:
Making Home CCTV Data Practices Visible

A1.2 Interim Project Critique

DECO7380

Team: ANN

Behind the Lens:

Making Home CCTV Data Practices Visible

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1 Table of Contents

Table of Contents **1**

1. Overview **3**

1.1 Project Background **3**

1.2 Problem Statement **3**

1.3 Project Development Since Previous Checkpoint **4**

2. Research Foundation **4**

2.1 Background Reading and Literature Review **4**

2.2 User Research **5**

2.3 Stakeholder Analysis **5**

2.4 Key Findings **6**

3. Concept Development **8**

3.1 From Research Findings to User Needs **8**

3.2 Design Rationale **9**

3.3 Conceptual Model **10**

3.4 Privacy Information and Design Considerations **12**

3.5 Ethical Considerations **14**

4. Prototyping / Sketching / Mock-ups **16**

4.1 Early Design Exploration **16**

4.2 Privacy Label — SEE **17**

4.2.1 Design Rationale 17

4.2.2 Exploring Privacy Label Representations 17

4.2.3 Exploratory Rating Framework 18

4.2.4 Current Privacy Label Prototype 20

4.2.5 Feedback and Questions 21

4.3 Privacy Guide — UNDERSTAND **21**

4.3.1 Design Rationale 21

4.3.2 Prototype Exploration 22

4.3.3 Feedback and Questions 22
4.4 Privacy Settings — CONTROL 23
4.4.1 Design Rationale 23
4.4.2 Prototype Exploration 23
4.4.3 Feedback and Questions 23
4.5 Access History — REVIEW 24
4.5.1 Design Rationale 24
4.5.2 Prototype Exploration 24
4.5.3 Feedback and Questions 25
5. Privacy Label Testing and Evaluation 25
5.1 Research Questions for Testing 25
5.2 Testing Method 27
5.3 Round1 - A/B/C Prototype Comparison Testing 28
5.4 Round2 - Multi-CCTV Comparison Testing 28
5.5 Testing Results 29
5.5.1 Round 1 — A/B/C Prototype Comparison 29
5.5.2 Round 2 — Multi-CCTV Comparison 29
5.6 Key Findings from Testing 30
5.7 Design Changes Based on Testing 30
6. Project Plan 31
6.1 Week 10 — Refine Prototype and Design User Flow 31
6.2 Week 11 — Prototype Integration and Internal Pilot 32
6.3 Week 12 — Final User Testing, Analysis and Iteration 32
6.4 Week 13 — Final Documentation, Exhibit and Presentation 32
6.5 Team Responsibilities 33
6.6 Timeline Summary 34
6.7 Project Risks and Management 35
7. Team Reflection 35

7.1 Progress and Changes 35

7.2 Team Collaboration 36

References 37

AI Use Statement 37

Appendices 37

Appendix A — Interview Questions 37

Appendix B — Participant Background 40

Appendix C — Interview Selected Quotes 40

Appendix D — Affinity Mapping 43

Appendix E — Finding–Evidence Matrix 43

Appendix F — Prototype A/B/C 45

Appendix G — Round 1 - Testing Raw Data 47

Appendix H — Muti_Level_Privacy_Label 49

Appendix I — Round 2 - Testing Raw Data 52

1. Overview

1.1 Project Background

Home CCTV systems are widely used for household security and everyday monitoring. However, connected CCTV systems involve more than video capture. Recordings may pass through local storage, cloud services, mobile applications, manufacturer infrastructure, and other connected services. Many of these processes are not directly visible to users.

This creates a privacy concern because CCTV cameras operate in private or semi-private environments and can capture sensitive information about people, activities, and daily routines. Previous research has identified privacy and security risks associated with smart cameras in sensitive environments (Wang et al., 2022), while smart-home research also shows that users may rely on manufacturers without fully understanding how device data practices operate (Zheng et al., 2018).

These concerns also affect people beyond the main account owner. Household members, visitors, and other bystanders may be captured without being involved in configuring or managing the CCTV system. Yao et al., (2019) show that smart-home bystanders may have different privacy expectations and less control than device owners.

Therefore, this project investigates how interaction design can make home CCTV privacy practices more **visible, understandable, and actionable**.

1.2 Problem Statement

Our previous research found that users could operate CCTV systems without clearly understanding what happened to their recordings behind the interface. Participants generally understood visible elements such as the camera and mobile application, but were less certain about data transmission, storage, manufacturer involvement, and access conditions.

This reflects research showing that users can have incomplete mental models of data flows between connected smart devices (Windl et al., 2024). The problem is also not simply that privacy information is unavailable. CCTV systems may already provide policies, settings, encryption options, storage controls, and user-management functions, but users may still struggle to locate, interpret, and connect this information to meaningful action. Feng et al., (2021) similarly argue that privacy choices need to be understandable and actionable.

Providing more information is therefore not automatically the solution. Caven et al., (2024) found that the presentation of IoT security information affects comprehension and cognitive effort, suggesting that transparency must consider both **what information is shown and how it is presented**.

The project therefore defines the core problem as:

> **Home CCTV users may have access to privacy information and controls, but they do not always have a clear and connected way to understand what happens to their recordings, who may access or process them, and what actions they can take.**

The current design direction addresses this through four connected stages: **SEE → UNDERSTAND → CONTROL → REVIEW**

These stages form the conceptual model developed later in the report.

1.3 Project Development Since Previous Checkpoint

Since the previous checkpoint, the project has progressed through additional user research, design exploration, concept integration, and low-fidelity prototype testing.

First, we conducted **seven additional interviews** to strengthen the user evidence and examine how people understand CCTV data practices, privacy information, manufacturer access, and user control. The new interview data was reviewed together with our previous research, leading to an updated set of **six key findings (F1–F6)**. These findings provided a clearer understanding of users' difficulties in understanding CCTV data flows, access conditions, privacy information, and available controls.

Based on this updated evidence, the project scope was refined. Rather than treating home CCTV privacy as a broad privacy problem, we narrowed the focus to two connected issues: **privacy transparency and meaningful user control**. We then explored four related design directions: **Privacy Label, Privacy Guide, Privacy Settings, and Access History**. These were initially developed as separate design explorations, but further analysis showed that they could be connected as part of one privacy journey. This led to the current integrated concept: **SEE → UNDERSTAND → CONTROL → REVIEW**

Finally, we began **low-fidelity prototype testing**, starting with the Privacy Label. Different information representations were tested to investigate how users understand, compare, and interpret CCTV privacy information. The findings from this testing are now being used to refine the wider integrated concept and guide the next design iteration.

2. Research Foundation

2.1 Background Reading and Literature Review

Existing research suggests that privacy challenges in smart-home systems are not only caused by technical risks, but also by users' limited understanding of how their data is collected, stored and accessed. Users may have incomplete mental models of the wider data ecosystem surrounding connected smart-home devices, including where information is processed and who may have access to it(Abdi et al., 2019) This is particularly relevant to CCTV systems, where recordings may involve local storage, cloud services, mobile applications and manufacturer systems.

More structured approaches have therefore been explored to make privacy and security information easier for users to understand. Consumers may struggle to locate or interpret privacy information when purchasing IoT devices, while simplified privacy and security labels can make this information more accessible(Emami-Naeini et al., 2019). Privacy information can also be organised into different layers, allowing important

information to be shown first while additional details remain available when needed(Emami-Naeini et al., 2020).

However, providing more information does not necessarily improve understanding. The amount and organisation of information presented in IoT privacy and security labels can influence user comprehension and decision-making(Chen et al., 2024). Similarly, increasing transparency through privacy dashboards does not always lead users to change their privacy settings or behaviour(Farke et al., 2021).

Together, these studies suggest that CCTV privacy information should not simply provide more technical detail. Instead, information should be prioritised, understandable and progressively disclosed, particularly for important issues such as storage, retention and data access. These ideas provided a foundation for the user research conducted in this project.

2.2 User Research

User research was conducted to investigate how users understand and manage privacy in home CCTV systems. Additional interviews were conducted after the previous checkpoint and analysed together with the earlier interview data. Affinity mapping was used to identify repeated concerns, differences and patterns across participants.

The interviews showed that participants often had an incomplete understanding of how CCTV recordings were stored and moved through the system. Some were unsure whether recordings were stored locally, in the cloud, or in both locations, and many were unclear about retention periods. Participants generally understood access by themselves or household members more easily than access by manufacturers, technical-support staff or other external parties.

Another repeated issue concerned the presentation of privacy information. Many participants reported skipping or only partly reading privacy information because it was long, complex or contained too much legal language. Instead, participants commonly preferred short and easy-to-scan information showing key details such as storage, access, retention and deletion. Participants also wanted privacy information to be connected to useful controls, including managing shared access, deleting recordings, reviewing access history and receiving access notifications.

2.3 Stakeholder Analysis

The project involves several stakeholder groups with different relationships to CCTV data. CCTV owners are the primary users because they configure devices, manage recordings and control privacy settings. Household members and shared users may also access recordings or appear in recorded footage, but often have less control over system settings. Visitors, neighbours and other bystanders may be captured by cameras without directly interacting with the system, meaning that decisions about recording, retention and sharing can also affect their privacy.

Other important stakeholders include CCTV manufacturers, cloud-service providers, technical-support staff and installers, because they influence how recordings are stored, processed, accessed and explained to users. The project has therefore considered not only the perspective of device owners, but also how external access, support access and shared access should be communicated.

This broader stakeholder perspective showed that privacy transparency is not only about giving CCTV owners more information. It also requires clear boundaries around who can access recordings, under what conditions, and what level of control different stakeholders have. These considerations contributed directly to the project's later findings and design requirements.

2.4 Key Findings

The combined interview, stakeholder and literature evidence resulted in six key findings that define the main privacy challenges addressed by the project.

F1 — Users have an incomplete understanding of CCTV data storage and data flow.

Many participants were unsure about where CCTV recordings were stored or how recordings moved between the camera, cloud service, and mobile application. For example, P2 knew that recordings could be viewed through the app but did not understand how the video reached the app. P6 was unsure whether recordings were stored locally, in the cloud, or in both places. This was also reflected in the Mental Model Activity, where participants' drawings showed gaps or uncertainty in the data journey between the camera and the user's device. This suggests that users can operate CCTV systems without fully understanding what happens behind the interface. Similar problems have been found in smart-home research, where users can have incomplete mental models of wider data systems(Abdi et al., 2019). Privacy and security information about IoT devices can also be difficult for consumers to find(Emami-Naeini et al., 2019).

Design implication: Information about storage, retention and data handling should be visible and explained in simple language.

F2 — Manufacturer and external access conditions are unclear to users.

Participants understood their own or household access better than manufacturer or external access. Several did not know whether manufacturer employees could view recordings or under what conditions this could happen. Participants also expected technical-support access to have a clear reason, permission, or notification. This shows that users need to understand not only who can access their data, but also why and when access may happen. Access and data-sharing practices are also important to how consumers understand privacy and security risks in IoT products(Emami-Naeini et al., 2020).

Design implication: Users should be able to see who can access CCTV data, why access may occur, and whether permission is required.

F3 — Unauthorised access and data incidents can damage user trust.

Participants repeatedly connected privacy problems with trust. P1 stated that manufacturer access without permission would cause a loss of trust, while P5 identified data leaks as a major concern. Other participants also indicated that serious privacy incidents could affect whether they continued using a product. This suggests that users care about both what access is possible and what has actually happened to their data. Data sharing, retention, and other privacy practices can also affect perceived risk and consumers' responses to IoT products(Emami-Naeini et al., 2019).

Design implication: Important privacy-related access and data events should be visible so users can understand how their data has been handled.

F4 — Long and complex privacy information discourages engagement.

Many Many participants reported that they skipped privacy information, read only part of it, or accepted it during setup without fully reading it. P6 explained that the information was long and contained too many legal words. P7 and P8 also reported that they did not fully read the privacy information. However, this was not true for every participant. P5 had read privacy information, showing that engagement can differ between users. The interviews suggest that the problem is not simply that users do not care about privacy. Participants were interested in issues such as access, storage, and deletion, but they did not always want to read long documents to find this information. The way privacy information is presented can therefore affect whether users engage with it.

Previous research also shows that information-heavy IoT labels can require more time and mental effort. The amount and presentation of information can affect comprehension and cognitive load(Caven et al., 2024).

Design implication: Essential privacy information should be short and easy to scan. More detailed information should remain available through progressive disclosure.

F5 — Users want key privacy information to be easy to find and understand.

Participants repeatedly wanted clearer information about where recordings were stored, who could access them, how long they were retained, and what happened to their data. P3 wanted to know “who can see the video” and “where it’s stored.” P7 preferred a short privacy summary that was visual and easy to scan. P9 also wanted a simple page showing storage, retention, access, and deletion information. These responses show that participants were not asking for every privacy detail to be shown at the same time. Instead, they wanted the most important information to be visible first. More detailed information could then be available when they wanted to learn more.

This approach is also supported by previous IoT privacy-label research. Structured and layered labels can make important privacy information easier to access while keeping further details available when needed(Emami-Naeini et al., 2019, 2020). Information presentation can also affect usability and cognitive load(Caven et al., 2024).

Design implication: Key privacy information should be presented in a clear and easy-to-find summary, with more detailed information available when needed.

F6 — Privacy transparency should be connected to meaningful user control.

Participants wanted not only to understand CCTV data practices but also to take action. They wanted controls for shared access, deletion, storage, access history, notifications, and permission. P7 stated that manufacturers should “give users more control and be clear about what’s happening.” This shows that understanding privacy information is more useful when users can also act on it. Privacy choices in IoT systems need to be understandable and actionable, and access control can affect how users evaluate privacy risks(Feng et al., 2021).

Design implication: Privacy information should connect to relevant settings and actions so users can manage the privacy controls available to them.

Together, these findings shifted the project away from treating CCTV privacy as a broad problem. The current focus is on two connected needs: making privacy practices transparent and giving users meaningful control over them. These findings directly informed the design requirements and conceptual model discussed in Section 3.

3. Concept Development

3.1 From Research Findings to User Needs

The six findings identified through the literature review and user research were synthesised to identify the main privacy needs that should guide the next stage of design development. Rather than responding to each finding separately, related findings were grouped according to the type of difficulty users experienced: visibility, understanding, action and later review.

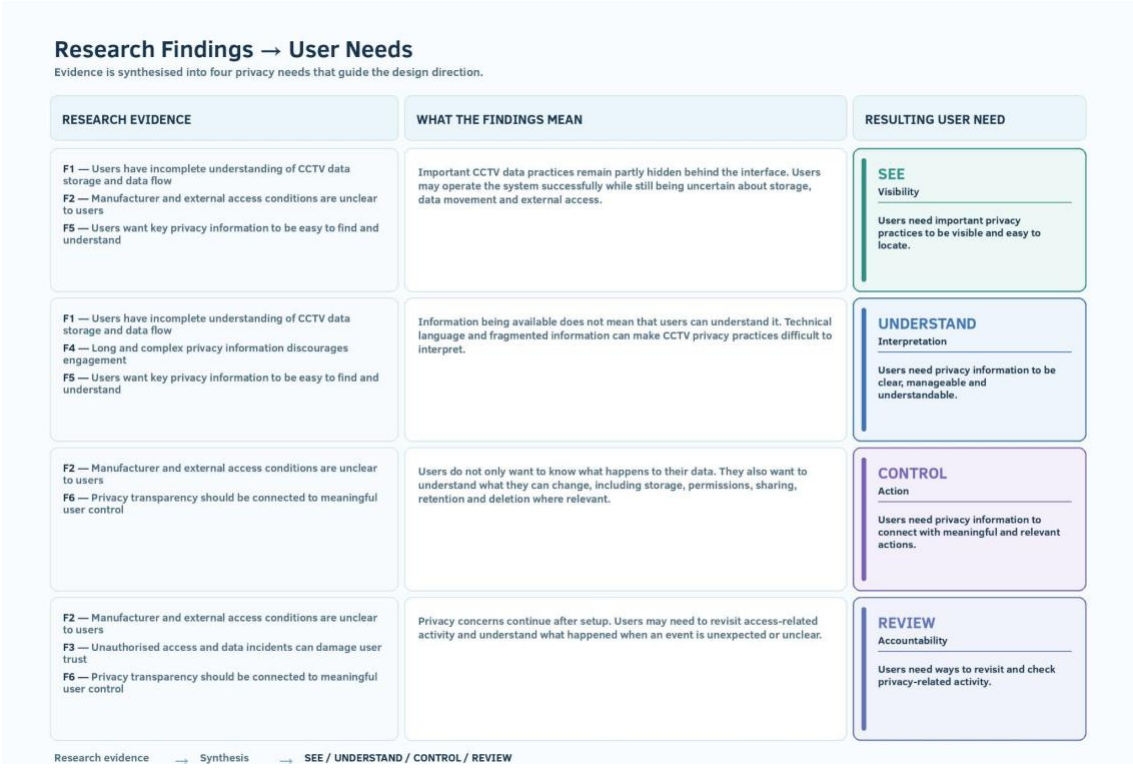


Figure 1. Research findings synthesised into four connected user needs: SEE, UNDERSTAND, CONTROL and REVIEW.

These four needs do not represent four separate privacy problems. Instead, they describe different forms of support that may be required across the CCTV privacy journey. Together, they narrowed the project from a broad concern with home CCTV privacy to a more specific design focus on privacy transparency and meaningful user control. The following section explains why these needs were developed into an integrated design direction.

3.2 Design Rationale

The four user needs do not occur at the same moment. Privacy decisions can happen before purchasing a CCTV product, during setup, during everyday use, and after a user wants to review an event. For this reason, the project does not treat privacy transparency as a single screen or one-time privacy notice.

The first design rationale is that **privacy information should be available at the stage where it is useful**. Before purchase, users may need a concise way to compare important privacy practices. During setup or later learning, they may need explanations of how data moves through the system. During everyday use, they may need access to current privacy settings and controls. After an event, they may need information that helps them review relevant access activity.

The second rationale is that **information alone is not enough**. Our user research showed that privacy information may already exist but can still be difficult to locate, interpret or connect to action. Feng et al., (2021) similarly argue that meaningful privacy control depends on users being able to understand and act on privacy choices.

Therefore, the design direction connects transparency with understanding and control rather than treating disclosure as the final goal.

The third rationale is that the four earlier design explorations should not be treated as independent solutions. The Privacy Label, Privacy Guide, Privacy Settings and Access History were initially explored separately, but they respond to different parts of the same privacy journey. Integrating them provides clearer roles for each concept and reduces unnecessary overlap.

Finally, the project aims to **support user judgement rather than provide one universal privacy judgement**. Different households may value storage, convenience, remote access and control differently. A cloud-based system is not automatically unsafe, and local storage is not automatically private. The design should therefore make relevant differences visible without reducing privacy to one overall “good” or “bad” result.

These principles led to the integrated direction:

SEE → UNDERSTAND → CONTROL → REVIEW

3.3 Conceptual Model

Privacy is not only the topic of this project; it also creates constraints on how the proposed design should work. The aim of the integrated concept is to improve transparency and user control, but these goals can create new risks if they are implemented without considering the different people affected by CCTV systems.

The project therefore treats ethical considerations as **design constraints rather than a separate checklist**. Each ethical issue is linked to a possible harm and a corresponding design response. This helps the team decide not only what the system should provide, but also what information or functionality should be limited, qualified, or avoided.

Research on smart-home bystanders shows that people who are recorded may have different privacy expectations and lower levels of control than the device owner (Yao et al., 2019). This is particularly important for home CCTV because one account holder may control the system while multiple household members, visitors, neighbours, or other bystanders are affected by the recording.

From Findings to Design Direction

Research Evidence	Identified Need	Design Response
F2, F4, F5	Key privacy information needs to be visible and easy to scan	**SEE — Privacy Label**
F1, F2	Users need help understanding data flow and access conditions	**UNDERSTAND — Privacy Guide**
F6	Users need to act on privacy information	**CONTROL — Privacy Settings**

F2, F3, F6	Users need to know what access actually occurred	**REVIEW — Access History**

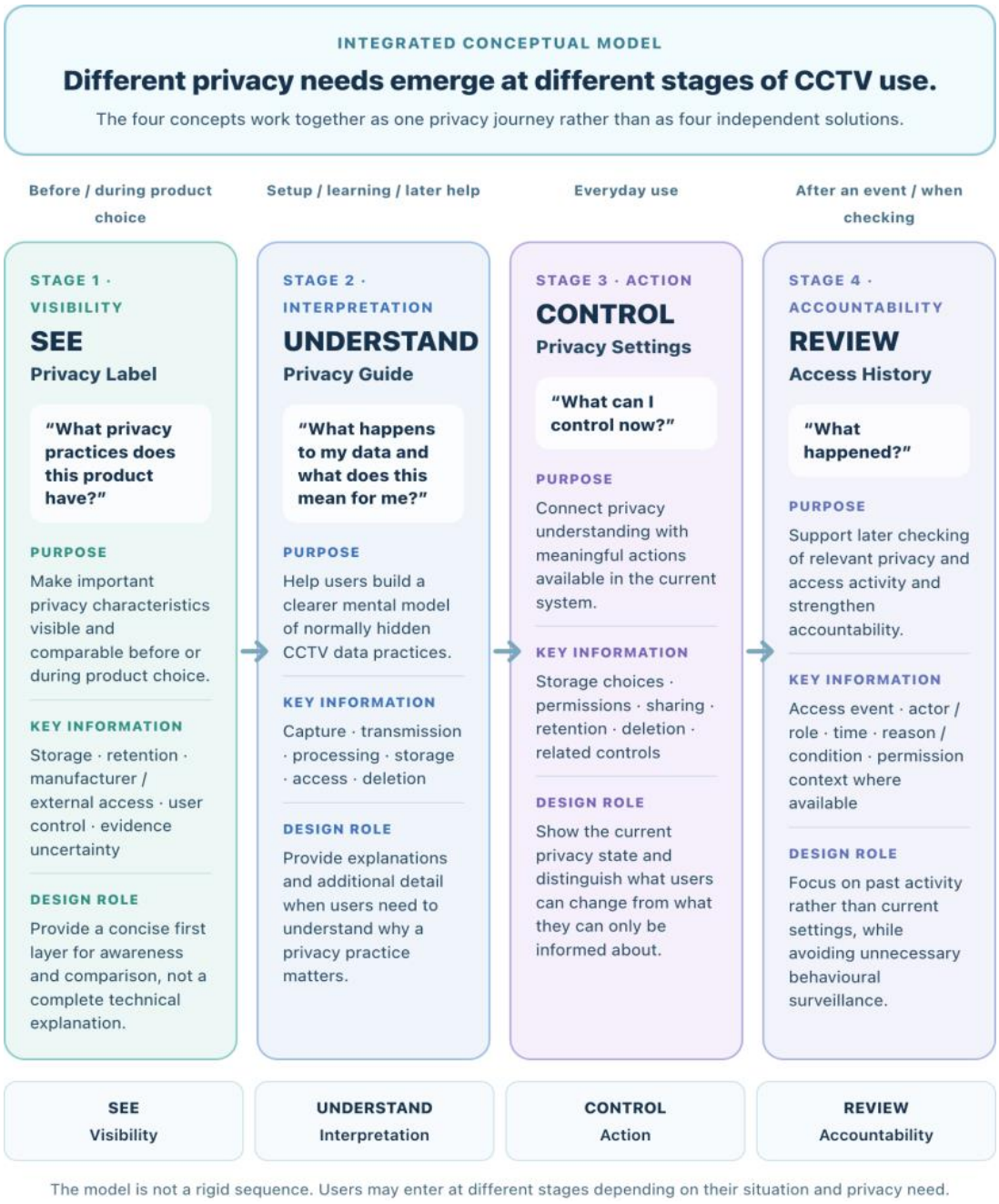


Figure 2. Integrated conceptual model showing the privacy journey from SEE and UNDERSTAND to CONTROL and REVIEW.

The first ethical constraint concerns **whose privacy the system represents**. The account owner usually has the greatest technical control, but this does not mean that other people affected by the camera have the same preferences or have consented to the same data practices. Yao et al., (2019) demonstrates that owners and bystanders can have different privacy expectations. For this reason, the conceptual model should not treat the owner's perspective as the only relevant perspective.

A second issue is the relationship between **transparency and surveillance**. Features such as Access History can improve accountability, but providing increasingly detailed information about other household users could also create new opportunities for monitoring. The REVIEW stage should therefore focus on information required to understand privacy-relevant access, rather than becoming a detailed record of another person's behaviour. This distinction is particularly important when deciding what information should appear in future Access History prototypes.

Another ethical concern is **false transparency**. Simplifying privacy information through labels and ratings can improve accessibility, but it can also make incomplete evidence appear certain. The current design therefore distinguishes between confirmed practices and information that is unknown or not disclosed. A missing manufacturer statement should not automatically produce a safer score. Similarly, Data Access Exposure and User Control should remain separate dimensions rather than being collapsed into one overall privacy grade.

The amount of information presented is also an ethical design issue because an interface can technically disclose information while still making it difficult for users to understand. The project therefore uses progressive disclosure: essential information is shown first, while explanations and more detailed evidence remain available when needed. This approach is consistent with layered IoT privacy information proposed by Emami-Naeini et al., (2020) and with research showing that information structure can influence comprehension and cognitive effort (Caven et al., 2024).

Finally, the project aims to protect **user autonomy**. Privacy requirements differ between households, and users may make different trade-offs between cloud services, remote access, storage, cost, convenience, and privacy. The role of the design is therefore not to tell users which CCTV product is universally best. Instead, it should make important practices and trade-offs understandable enough for users to make and explain their own decisions.

These considerations establish three broader ethical boundaries for the project:

> **Transparency should not become surveillance.**
> **Simplification should not create false certainty.**
> **Decision support should not replace user judgement.**

These boundaries influence the design of all four stages of the conceptual model and will continue to guide the prototype development and evaluation described in the following sections.

3.4 Privacy Information and Design Considerations

The conceptual model defines **when** different forms of privacy support may be useful, but it does not by itself determine **how privacy information should be represented**. Across SEE, UNDERSTAND, CONTROL and REVIEW, the project therefore applies a shared set of information-design considerations.

Rather than creating separate rules for each prototype, these considerations are organised around three priorities: privacy information should be **clear, trustworthy and actionable**. These priorities help maintain consistency across the four stages while still allowing each concept to serve a different purpose.



Figure 3. Shared information-design considerations across the four stages: clarity, trustworthiness and actionability.

The first consideration is clarity. Privacy practices are often conditional rather than binary. For example, manufacturer access may depend on technical support, account configuration, user permission or other conditions. Presenting this simply as “access: yes” or “access: no” could hide the information users actually need. Similarly, storage, retention and deletion should not be treated as interchangeable concepts. A system may use cloud storage but retain recordings for different periods, while deletion conditions may introduce another part of the data lifecycle. The project therefore aims to show these relationships without collapsing them into one simplified statement.

Clarity also requires managing the amount of information presented. The project follows a **progressive disclosure** approach in which essential information is presented first and supporting detail is available when required. Layered approaches have previously been proposed for IoT privacy and security information (Emami-Naeini et al., 2020), while Chen et al., (2024) show that the complexity and organisation of IoT privacy information can affect user comprehension and behaviour. The aim is therefore not to remove complexity, but to make it manageable.

The second consideration is **trustworthiness**. Data Access Exposure and User Control represent different aspects of privacy and should remain distinguishable. A system may involve more external data processing while still providing strong deletion, storage or permission controls. Combining these dimensions too early into one overall score could hide these differences. The design also avoids treating unavailable evidence as a positive privacy outcome. Where a practice cannot be verified from the available evidence, the interface should communicate that uncertainty rather than automatically classifying it as safe.

The third consideration is **actionability**. Privacy information becomes more useful when users can understand what action follows from it. Feng et al., (2021) argue that meaningful privacy choices require users to understand the choices available to them. In this project, information about storage, permissions, sharing, retention or deletion should therefore connect to relevant controls where such controls exist. However, the design should also clearly identify situations where a user can only be informed about a practice rather than change it.

Overall, these principles create a consistent information structure across the conceptual model. **SEE** prioritises clear comparison, **UNDERSTAND** adds explanation, **CONTROL** connects information with present actions, and **REVIEW** supports later accountability. The next section considers the ethical limits that further constrain how these principles should be applied.

3.5 Ethical Considerations

Privacy is not only the topic of this project; it also creates constraints on how the proposed design should work. The aim of the integrated concept is to improve transparency and user control, but these goals can create new risks if they are implemented without considering the different people affected by CCTV systems.

The project therefore treats ethical considerations as **design constraints rather than a separate checklist**. Each ethical issue is linked to a possible harm and a corresponding design response. This helps the team decide not only what the system should provide, but also what information or functionality should be limited, qualified, or avoided.

Research on smart-home bystanders shows that people who are recorded may have different privacy expectations and lower levels of control than the device owner (Yao et al., 2019). This is particularly important for home CCTV because one account holder may control the system while multiple household members, visitors, neighbours, or other bystanders are affected by the recording.



Figure 4. Ethical design considerations and constraints guiding the proposed CCTV privacy system.

The first constraint is that **greater transparency for the account owner should not create greater surveillance of other people**. This is especially important for CONTROL and REVIEW. Privacy Settings may clarify roles and permissions, but Access History should not become a detailed behavioural record of household members. The design therefore focuses on information needed for privacy accountability rather than unnecessary monitoring.

The second constraint concerns **evidence and uncertainty**. Simplified labels and ratings can improve accessibility, but they may also appear more certain than the available evidence allows. The project therefore keeps unknown or not-disclosed information visible rather than treating missing evidence as safe. At the same time, detailed evidence should not overwhelm the first layer of information. Progressive disclosure is used so that users can access more detail when required(Emami-Naeini et al., 2020) .

The third constraint is **user autonomy**. The project does not assume that there is one universally best CCTV privacy configuration. Different users may make different trade-offs between privacy, remote access, storage, convenience and cost. The role of the design is therefore to make these differences understandable and support informed decisions rather than determine the decision for the user.

Together, these constraints establish three boundaries for the project:

> **Transparency should not become surveillance.**
> **Simplification should not create false certainty.**
> **Decision support should not replace user judgement.**

4. Prototyping / Sketching / Mock-ups

4.1 Early Design Exploration

The updated research findings showed that the privacy problem could not be addressed only by giving users more information. Participants had difficulty understanding how CCTV data moved through the system (F1), were uncertain about manufacturer and external access (F2), and often avoided long or complex privacy information (F4). At the same time, participants wanted important privacy information to be easier to find and understand (F5). They also wanted clearer ways to manage their privacy after understanding the information (F6). These findings led us to explore privacy as a process rather than a single screen or feature. The early design direction was organised around four user needs:

SEE → UNDERSTAND → CONTROL → REVIEW

- **SEE:** make important privacy practices visible.
- **UNDERSTAND:** explain how CCTV data is handled.
- **CONTROL:** connect privacy information to actions users can take.
- **REVIEW:** allow users to check what has happened to their data.

The four components were therefore not developed as four separate final solutions. They were used to explore different parts of one privacy experience. At this stage, we used low-fidelity prototypes. The main purpose was to explore the information structure, user understanding, and interaction flow before developing a more detailed interface.

4.2 Privacy Label — SEE

4.2.1 Design Rationale

The Privacy Label was developed in response to F2, F4 and F5. Participants wanted to know basic privacy information such as where recordings were stored, who could access them and how long they were retained. However, long privacy policies and technical descriptions were difficult to engage with.

This problem is also reflected in previous research. Emami-Naeini et al., (2019) found that privacy and security information can be difficult for consumers to locate when considering IoT products. Later research explored how important IoT privacy and security information could be organised in a more structured label(Emami-Naeini et al., 2020). Privacy notice research also suggests that the timing, presentation and amount of information influence whether privacy information is useful to users(Schaub et al., 2015).

However, simplifying information also creates a design challenge. Chen et al., (2024) and Caven et al., (2024) show that the amount and presentation of IoT privacy and security information can affect comprehension and decision-making. Therefore, our goal was not only to make the label shorter. We also wanted to explore how key information could be made easy to see without removing details that users may need.

4.2.2 Exploring Privacy Label Representations

The interview findings helped identify **what privacy information should be included**, but they did not provide a clear answer about **how this information should be presented**. F4 showed that long and complex privacy information can discourage engagement, while F5 showed that participants wanted key information to be concise, easy to find and understandable. This created a design question: **how can the same CCTV privacy information be presented in a form that users can quickly understand and compare?**

This question is also supported by previous privacy-label research. Emami-Naeini et al., (2020) explored structured privacy and security labels for IoT devices, while Schaub et al., (2015) showed that the presentation and amount of privacy information can affect whether privacy notices are useful to users. Caven further found that adding more information does not necessarily improve decision accuracy and can increase decision time and cognitive load (Caven et al., 2024) . These studies supported the use of a structured label, but did not determine which representation would work best for our CCTV context.

We therefore used alternative prototypes to explore the **presentation of the same underlying privacy information**. We initially developed two approaches and later added a third representation to broaden the comparison:

- **Prototype A — Facts Only:** presented the key privacy information directly as short factual statements.

- **Prototype B — Rating Label:** retained the factual information but added summary ratings to explore whether an additional overview could support faster interpretation and comparison.
- **Prototype C — Visual Profile:** presented the privacy dimensions more visually to explore whether users could understand the overall privacy profile with less reliance on text.

The privacy information represented across the three prototypes was kept consistent. The main variable was therefore the **form of representation**, rather than the underlying product information. This allowed the prototypes to act as a controlled design exploration of facts, ratings and visual summaries. The purpose of the comparison was not simply to ask participants which prototype they liked most. Preference was considered together with task performance, interpretation, confidence, reading behaviour and misunderstanding. The results were then used to identify which elements should be **retained, revised or removed** in the next Privacy Label iteration.

Prototype B introduced an additional design question because its summary ratings required a consistent way to translate detailed privacy information into broader dimensions and levels. The development and limitations of this exploratory rating framework are described in Section 4.2.3.

4.2.3 Exploratory Rating Framework

Prototype B introduced summary ratings as an additional layer above the factual privacy information. The purpose of the rating was to explore whether users could use a simple summary to form an initial understanding of a CCTV product before checking the detailed information underneath. The rating was not intended to replace the privacy facts or provide an overall judgement of whether a product was “good” or “bad” for privacy. Instead, it was designed as a **secondary reference point** for comparison. The underlying facts remained visible so that users could understand what information contributed to the summary.

From Privacy Information to Rating Dimensions:

The rating framework was based on the same privacy information used across the three label prototypes. Rather than producing one overall privacy score, related privacy factors were grouped into two dimensions: **Data Access Exposure** and **User Control**.

Data Access Exposure represents the extent to which CCTV recordings may be accessed, processed or shared beyond the user's direct control.

User Control represents the extent to which users have meaningful options to manage relevant privacy practices.

These dimensions were kept separate because they represent different aspects of privacy. A CCTV system may involve greater external data processing while still providing users with relatively strong privacy controls. Combining them into one overall score could therefore hide important differences between products.

Indicator-Based Scoring Method:

To construct the summary ratings consistently, each dimension was divided into **five indicators**. Each indicator was assigned a score of **0, 1 or 2** based on the available product evidence.

For **Data Access Exposure**, a higher sub-score represents greater exposure:

Sub-score	Meaning
0	Lower or limited exposure
1	Moderate or conditional exposure
2	Broader exposure

For **User Control**, a higher sub-score represents stronger control:

Sub-score	Meaning
0	Little or no meaningful user control
1	Partial or conditional user control
2	Stronger direct user control

The five indicator scores were then added together:

Final dimension score = Indicator 1 + Indicator 2 + Indicator 3 + Indicator 4 + Indicator 5

As each indicator could receive a maximum score of 2, each dimension had a maximum total of **10 points**:

5 indicators × 2 points = 10 points

The final result was therefore presented as a score out of 10, such as **6/10 for Data Access Exposure** or **7/10 for User Control**. Importantly, the two scores have different directions. For **Data Access Exposure**, a higher score means greater exposure. For **User Control**, a higher score means stronger user control. The two dimensions were therefore intended to be interpreted separately rather than combined into one overall privacy score. The 0–2 scoring method was used to create a simple and consistent way of translating the underlying privacy evidence into the prototype. It was not intended to suggest that privacy could be measured precisely. The final 10-point score was the sum of the five indicator assessments, rather than an independent numerical judgement made by the design team.

Representing Uncertainty:

Not all CCTV privacy information could be assessed with the same level of certainty. Manufacturer documentation could be incomplete, unclear or dependent on particular settings, subscriptions or conditions. Missing evidence was therefore not automatically treated as either a positive or negative privacy condition. Where one or more indicators could not be assessed confidently, uncertainty could be represented through a **range or question mark** rather than presenting one definite score. A range therefore represented **uncertainty in the available evidence**, rather than variation in product performance. This followed an important principle of the project: **simplification should not create false certainty**.

Role and Limitations of the Framework:

This scoring framework was developed as an **operational method for constructing Prototype B**, rather than as a validated measurement of CCTV privacy. Its purpose was to make the rating representation consistent enough to investigate a design question: **can a simplified score help users understand and compare privacy information while the supporting facts remain available for verification?** The formative evaluation therefore examined whether participants understood the two dimensions, the direction of the scores, and the representation of uncertainty, as well as how they used the scores alongside factual information. These findings were then used to revise the Privacy Label rather than to validate the scoring framework itself.

4.2.4 Current Privacy Label Prototype

Based on the prototype exploration and formative testing, the current Privacy Label combines the **quick overview provided by Prototype B** with clearer factual information and additional explanation where needed.

The current prototype includes five main design elements:

1. **Summary ratings** — Data Access Exposure and User Control provide a quick overview of two different privacy dimensions. Their directions are explicitly explained because testing showed that users could otherwise interpret a higher score as automatically better.
2. **Key privacy facts** — information such as storage, manufacturer access and user control remains directly visible. These facts are not replaced by the ratings, as participants repeatedly used concrete information when explaining their understanding and comparison of CCTV products.
3. **Clear rating direction** — the interface explains that higher **Data Access Exposure** means greater exposure, while higher **User Control** means stronger control. This was added to address confusion identified during testing.
4. **Visible uncertainty** — where product evidence is incomplete or conditional, the label communicates this uncertainty rather than presenting a definite result. Additional explanation is provided to clarify that a range or unknown value reflects limitations in the available evidence.

5. **Progressive disclosure of supporting evidence** — the main label prioritises information needed for quick scanning, while more detailed explanations and supporting evidence are available as a secondary layer. This allows users to inspect how a rating was formed without placing all technical information on the first screen.

Together, these changes position the Privacy Label as an entry point to the wider privacy experience. It supports the **SEE** stage by making important privacy information visible at a glance, while allowing users to move to more detailed information when further understanding is required.

4.2.5 Feedback and Questions

Two questions raised during the previous checkpoint helped clarify the role and limitations of the Privacy Label.

The first question concerned **manufacturer participation**: if a company benefits from unclear or exploitative data practices, why would it voluntarily use a transparency label? This feedback showed that the concept could not rely only on manufacturer goodwill. A meaningful label would require common disclosure rules, traceable evidence and some form of independent oversight or verification. At the current stage, however, the prototype does not claim to provide official certification. It is presented as a research-based representation compiled from available evidence. This question therefore shifted the concept from a simple information label towards a broader question of **accountability and evidence provenance**.

The second question was why consumers could not simply photograph a CCTV product and ask a chat-based tool to explain its privacy practices. A chat tool may provide useful additional explanation, but it requires users to recognise the privacy issue, know what to ask and judge whether the answer is complete and reliable. It may also present different information depending on the question asked. The Privacy Label therefore explores a different role: providing the same core privacy dimensions by default so that products can be compared consistently. A chat-based tool could still support later explanation, but it would not replace the standardised first layer of information.

These questions directly informed the later testing of information consistency, evidence status, score interpretation and uncertainty discussed in Section 5.

4.3 Privacy Guide — UNDERSTAND

4.3.1 Design Rationale

Making privacy information visible does not necessarily mean that users understand what happens behind it. F1 showed that participants often had incomplete mental models of CCTV data flow. Some knew that recordings could be viewed through an app, but were less certain about what happened between the camera, cloud services, manufacturer systems and their phone. Similar problems have been identified in smart-home research. Abdi et al., (2019) found that users can have incomplete understandings

of the wider systems involved in smart-home devices. Windl also showed that privacy concerns and mental models become more complex when data moves across interconnected devices and services(Windl et al., 2024). The Privacy Guide was therefore explored as a way to connect individual privacy facts to the wider system. Its purpose is not to reproduce the full privacy policy. Instead, it provides additional explanation when users want to understand what a privacy statement means or how a particular data practice works.

4.3.2 Prototype Exploration

The guide focuses on questions that appeared repeatedly during the research:

- Where does my recording go?
- Is it stored locally or in the cloud?
- Who may access it?
- Why might a manufacturer or external service need access?
- What part of this process can I control?

The prototype uses a simplified data journey to show the relationship between the camera, storage, cloud or manufacturer services, and the user application. This component also supports progressive disclosure. The Privacy Label provides the first layer of information, while the Privacy Guide provides further explanation when required. This avoids placing all technical information on the first screen while still allowing users to investigate the details behind a privacy statement.

4.3.3 Feedback and Questions

Checkpoint feedback asked whether the project was attempting to create an application that would guide users through the CCTV data flow, and how users could know whether such guidance was trustworthy.

This question helped clarify that the Privacy Guide should not become a separate source of unsupported privacy claims. Its role is to help users **interpret information that is already supported by product documentation or other available evidence**. For example, the guide may explain the relationship between the camera, local storage, cloud services and the mobile application, but it should distinguish between documented information and uncertain or unknown parts of the data flow.

This feedback also strengthened the connection between the Privacy Label and Privacy Guide. The Label provides a concise first layer, while the Guide explains the meaning and context behind individual privacy statements. Trust should therefore come from **traceable evidence and clear explanation**, rather than from asking users to trust the interface itself.

A remaining design question is how much technical detail users need before additional explanation becomes information overload. This will require further prototype testing.

4.4 Privacy Settings — CONTROL

4.4.1 Design Rationale

The interviews suggested that transparency alone was not enough. F6 showed that participants also wanted to know what they could do after learning about a privacy issue. Examples included changing shared access, managing recording settings, deleting recordings and controlling storage.

This distinction between information and action is important. Feng et al., (2021) argue that meaningful privacy control in IoT systems depends on how privacy choices are presented and made available to users. Farke et al., (2021) also found that providing privacy information through a dashboard does not automatically lead users to change their privacy settings or behaviour.

The design direction therefore connects privacy information with relevant controls instead of treating the Privacy Label and Privacy Settings as unrelated parts of the system.

4.4.2 Prototype Exploration

The Privacy Settings component explored controls related to the issues identified in the interviews, including:

- shared-user access;
- recording and storage options;
- retention and deletion;
- camera and microphone settings; and
- data-sharing choices.

The intended interaction is that users can move from understanding a privacy practice to the relevant setting when an action is available.

For example, if the Privacy Label shows that recordings are shared with authorised household users, the user should be able to move to the relevant access setting rather than searching through unrelated menus.

The prototype therefore explores a direct relationship between **information and action**:
Privacy information → explanation → relevant control

This was intended to reduce the gap between knowing that a privacy practice exists and knowing what can actually be changed.

4.4.3 Feedback and Questions

Earlier feedback on the dashboard concept raised a potential conflict between **privacy protection and later access to important footage**. For example, if people captured by the camera were automatically blurred to protect bystander privacy, would important visual evidence still be available when a serious incident occurred?

This question showed that adding more privacy features does not automatically create a better privacy system. A feature designed to protect one person may reduce the usefulness of the recording in another context. Rather than treating automatic face blurring as a core feature of the current concept, the project narrowed CONTROL towards privacy settings that users can clearly understand and manage, including permissions, storage, retention, deletion and data-sharing choices.

The feedback therefore contributed to a broader design principle: **privacy controls should not hide their consequences from users**. If a future design includes masking or restricted visibility, it would need to explain what is hidden, whether the original footage still exists, who may access it and under what authorised conditions.

4.5 Access History — REVIEW

4.5.1 Design Rationale

The previous three components focus mainly on what the system says can happen. However, F2 and F3 showed that participants were also concerned about whether access had actually occurred. Manufacturer access, technical-support access and unauthorised access were particularly difficult for participants to understand because these activities are usually not visible.

This creates a difference between general transparency and event-level transparency. A privacy policy may state that access is possible under certain conditions, but this does not necessarily tell the user whether their own recordings have been accessed.

Research on privacy logs and accountability provides support for making these events more visible. Samavi & Consens, (2018) discuss the use of privacy logs to support transparency and accountability, while Urquhart et al., (2019)highlight the importance of making accountability more demonstrable within IoT systems.

4.5.2 Prototype Exploration

The Access History component was therefore explored as a record of privacy-relevant access events. Rather than only showing that access is technically possible, the design considers information such as:

- who accessed the recording;
- when the access occurred;
- the reason or purpose for access;
- whether permission was required; and
- whether the user had approved the access.

For example: **Technical Support — 18 September — Recording Access — Permission Approved**

The purpose of this component is not to show every technical system event. It focuses on access information that could help users understand and review how their CCTV data

has actually been handled. This extends the project from privacy awareness towards accountability.

4.5.3 Feedback and Questions

The earlier discussion about restricted or hidden footage also raised a wider accountability question: if access to CCTV data is limited for privacy reasons, how can users later understand what happened when access does occur?

This question contributed to the development of Access History as a separate REVIEW stage. Rather than showing every technical system event, the concept focuses on privacy-relevant access information, including who accessed the recording, when access occurred, why it occurred and whether permission was required.

However, greater logging also creates a new ethical concern. A highly detailed history could become another form of household surveillance if it records unnecessary information about other users. The current direction therefore limits Access History to information needed for privacy accountability rather than detailed behavioural monitoring.

A remaining question is what level of access detail is useful for accountability without creating additional surveillance. This issue will be explored in later design development.

5. Privacy Label Testing and Evaluation

5.1 Research Questions for Testing

The evaluation was developed from previous checkpoint feedback and unresolved questions in the Privacy Label prototype. Rather than testing whether participants simply “liked” the concept, the evaluation examined whether users could find, understand and compare CCTV privacy information, and whether the current representation created misunderstanding.

One question raised during the previous checkpoint was why consumers could not simply photograph a CCTV product and ask a chat-based tool about its privacy practices. This helped clarify the role of the Privacy Label. A chat-based tool requires users to know what to ask and evaluate whether the answer is complete and reliable. In contrast, the Privacy Label provides a consistent first layer of key information across products, such as manufacturer access, storage, retention and user controls. More detailed explanations can then be provided when needed. This follows research supporting appropriately timed and layered privacy information(Emami-Naeini et al., 2020; Schaub et al., 2015).

The checkpoint feedback also questioned how users could know whether the displayed information was accurate. The current label is therefore treated as a research-based representation of available public evidence, rather than an official certification system. Where evidence is incomplete or unclear, the design should communicate this uncertainty rather than present the information as equally certain.

Several additional questions emerged from the prototype. Data Access Exposure and User Control use opposite score directions, which may conflict with the common assumption that a higher score is always better. Score ranges and question marks intended to represent incomplete evidence may also be misunderstood. In addition, the current scoring method—five indicators scored from 0–2 and summed to a 10-point score—remains provisional rather than an established privacy standard. These issues therefore became important areas for testing.

Based on these uncertainties, six research questions were investigated across two rounds of formative testing.

Round 1 — Comparing Information Representations

RQ1: Can participants locate and accurately explain key information about manufacturer access, storage and user controls?

RQ2: What differences in comprehension, task time, reported confidence and preference are observed across the three presentation approaches?

Across Both Rounds — Interpreting the Rating

RQ3: How do participants interpret Data Access Exposure, User Control, score direction and uncertainty indicators?

Round 2 — Comparison, Evidence and Decision-Making

RQ4: Can participants use a consistent label structure to identify and explain privacy differences between CCTV products?

RQ5: Which privacy attributes do participants prioritise when comparing products, and what trade-offs do they describe?

RQ6: What changes in understanding, reported confidence and reported trust are observed after participants review the scoring guide and supporting evidence?

Together, these questions evaluate whether the Privacy Label supports accurate interpretation and comparison, whether its scoring and uncertainty mechanisms are understandable, and which elements should be retained, revised or removed in the next iteration.

5.2 Testing Method

A comparative task-based usability testing method combined with concurrent Think Aloud was used to examine how users interpreted, located and compared CCTV privacy information. The method captured both quantitative measures, including accuracy, completion time and confidence, and qualitative observations such as hesitation, misunderstanding and information-search behaviour.

In Round 1, the same Ring privacy information was presented in three formats—Facts Only, Rating Label and XY Privacy Profile—to compare different information representations. In Round 2, a consistent Privacy Label was applied across eufy, Ring and Tapo to examine cross-product comparison, score interpretation, privacy priorities and responses after reviewing the scoring guide and supporting evidence. Presentation order varied across records to reduce order and learning effects.

The analysis used coded records R1-P01–R1-P06 and R2-P01–R2-P06. These codes are used throughout the findings to connect observations and design decisions to individual records. Think Aloud may influence task completion time, so timing is treated as comparative evidence rather than ordinary shopping speed.

5.3 Round1 - A/B/C Prototype Comparison Testing

Round 1 compared three ways of presenting the same Ring Pan-Tilt Indoor Camera privacy information, keeping the underlying product information consistent so that differences in representation could be examined.

Prototype A – Facts Only presented privacy information as structured factual text without ratings. Prototype B – Rating Label added two research ratings—Data Access Exposure and User Control—alongside key privacy facts and rating explanations. Prototype C – XY Privacy Profile represented the same two dimensions on a two-dimensional graph to explore whether exposure and control could be understood visually.

For each prototype, participants answered the same questions about manufacturer access, user control and storage, followed by a confidence rating. Accuracy and completion time were recorded alongside observed attention, reading behaviour, hesitation, rereading and misinterpretation. Participants then compared the three designs for ease of understanding, information-finding speed and overall preference. Where recorded, presentation order varied across participants to reduce potential order effects.

This round primarily addressed RQ1–RQ3, examining information finding and comprehension, differences between presentation formats, and interpretation of the two privacy dimensions.

5.4 Round2 - Multi-CCTV Comparison Testing

Round 2 extended the evaluation from single-product comprehension to cross-product comparison and product-choice reasoning. Participants compared Privacy Labels for eufyCam S3 Pro, Ring Stick Up Cam Pro and Tapo C425. The three products used the same label structure, holding the presentation format relatively consistent so that differences in privacy practices could be compared more directly.

The evaluation used two stages. In Stage 1, participants reviewed the labels without the scoring guide or supporting evidence. They interpreted Data Access Exposure, User Control, score ranges and uncertainty indicators, compared manufacturer access, storage, retention and encryption, and made an initial product choice. In Stage 2, the

scoring guide and supporting evidence were introduced. Participants then re-explained the ratings and reconsidered their choices, allowing changes in understanding, confidence, reported trust and product-choice reasoning to be compared within the same record.

Recorded measures included comparison accuracy, completion time, reported confidence, product choice and observed interpretation difficulties such as hesitation, rereading and misunderstanding. Product order varied across records to limit potential order effects. This round primarily addressed RQ3–RQ6, examining score and uncertainty interpretation, cross-product comparison, privacy trade-offs, and how access to scoring rules and evidence influenced participants' interpretations and reasoning.

5.5 Testing Results

5.5.1 Round 1 — A/B/C Prototype Comparison

Six participants compared the three representations of the same Ring privacy information. Overall, Prototype B showed the strongest preference, with four participants ranking it first, while two preferred Prototype A and none selected Prototype C as their first choice.

Prototype A was valued for its direct factual presentation. Participants who preferred A generally found it straightforward because they could read the privacy facts without first interpreting a rating system. However, some participants considered the amount of text slower to scan.

Prototype B provided a faster visual overview through the Data Access Exposure and User Control ratings. Across the six records, its approximate average completion time was lower than both A and C. Participants also showed relatively high confidence when using B. However, the wording "High Exposure" was repeatedly confusing because a high value could initially be interpreted as a positive result. The letter grades and the reason behind individual scores also required additional explanation.

Prototype C was generally understood after participants read the axis labels, and several participants recognised its potential value for comparing multiple products. However, it received no first-place preference in the single-product comparison and usually required more interpretation than the other formats.

These results supported taking Prototype B forward, while retaining the factual clarity of Prototype A and revising how the ratings were explained.

5.5.2 Round 2 — Multi-CCTV Comparison

Six participants then tested the refined Privacy Label across eufyCam S3 Pro, Ring Stick Up Cam Pro and Tapo C425. The consistent structure allowed participants to locate and compare information such as Manufacturer Access, Video Storage, Video Retention, Encryption and User Control across the three products.

The large scores attracted attention quickly, but the two opposite score directions were not immediately clear to every participant. Some initially applied a conventional **"**higher is better" interpretation to both scores before rereading the explanation. Score ranges and question marks created greater difficulty. Participants commonly interpreted ranges as approximate ratings, setting-dependent values or changing product performance before reviewing the scoring guide.

After the scoring rules and evidence sources were shown, participants were better able to distinguish Data Access Exposure from User Control and understand that score ranges represented unresolved evidence rather than product variability. The additional evidence also generally increased confidence or trust in the comparison process.

Across the comparison tasks, Video Storage, Manufacturer Access and User Control repeatedly influenced participants' reasoning. Clear retention periods were also useful, particularly for less experienced CCTV users. In contrast, technical concepts such as E2EE, encryption and external processing often required more reading or explanation.

The results also showed that the label did not replace other purchasing considerations such as brand familiarity, price, usability or image quality. Instead, it introduced privacy-related information into the comparison and gave participants additional evidence for explaining their choices.

5.6 Key Findings from Testing

F1 Layered information and traceable facts.

R1-P01 challenged score credibility and dominance; R2-P01 valued matching factual headings and R2-P02 valued actionable choices. These support linking summaries to inspectable facts. The contrasting preferences in R1-P03 and R1-P04 provide additional cases for further testing.

F2 Score direction and visual interpretation.

R2-P02 initially treated both scores as higher-is-better, whereas R2-P01 explained them correctly. R1-P01 also needed clarification. The problem is inconsistent first-pass interpretation. R1-P05 and R1-P06 additionally flag letter-grade and XY-direction issues.

F3 Uncertainty and missing evidence.

R2-P02 confused missing evidence with configuration variation. R2-P01 broadly understood the range but requested identification of unconfirmed facts. R2-P03, R2-P04, R2-P05 and R2-P06 describe further confusion with approximation, settings, test variation and unavailable features.

F4 Individual priorities and trade-offs.

R2-P01 prioritised storage, permission and controls; R2-P02 initially relied on brand familiarity, retention and E2EE. R1-P01 stressed permission and R1-P02 challenged vague purposes. These differences favour comparable facts over a universal importance ranking.

F5 Understanding preference and confidence.

R2-P01 retained choice and confidence after clarification, while R2-P02 changed both. R1-P01 and R1-P02 show that preference and recorded speed can diverge. R2-P05 supplies an additional case in which clearer understanding does not produce greater reassurance.

F6 Access conditions and technical wording.

R1-P02's access questions and R2-P02's E2EE uncertainty show that naming encryption is insufficient. R1-P03, R1-P05 and R1-P06 flag defaults and access scope. R2-P05 and R2-P06 provide further cases concerning encryption and external processing.

5.7 Design Changes Based on Testing

The Privacy Label was revised based on repeated comprehension problems and user behaviours identified across both testing rounds, rather than preference alone.

1. Retain the layered structure of Prototype B.

Round 1 supported combining a quick visual summary with Key Privacy Facts and more detailed rating information. Prototype B was therefore taken forward as the main direction, while retaining the factual clarity of Prototype A.

2. Make score direction more explicit.

Because some participants assumed that higher scores were always better, isolated terms such as "HIGH" and letter grades were removed. The revised design uses clearer directional wording and visual cues to show that lower Data Access Exposure and higher User Control represent different meanings. Round 2 confirmed that these directions still require strong visual emphasis.

3. Explain uncertainty directly on the label.

Score ranges and question marks were repeatedly misinterpreted. Future versions will place a short explanation beside ranges, such as "Evidence range — some factors unknown", rather than requiring users to consult a separate guide first.

4. Prioritise concrete privacy facts.

Manufacturer Access, Video Storage and User Choices will remain visually prominent because these were repeatedly useful during product comparison. More technical terms such as E2EE will include short plain-language explanations

5. Keep detailed evidence as a secondary layer.

Scoring rules and source evidence increased trust but were rarely used during the first scan. They will therefore remain available through progressive disclosure rather than competing with the main privacy information.

6. Project Plan

During Weeks 10–13, the project will develop the current concepts into an integrated low- to medium-fidelity prototype. The remaining work follows four stages: **Design** → **Integrate** → **Evaluate** → **Deliver**, building directly on the research and testing completed so far.

6.1 Week 10 — Refine Prototype and Design User Flow

Week 10 will focus on refining the four components and defining how they connect across **SEE** → **UNDERSTAND** → **CONTROL** → **REVIEW**. The Privacy Label will be updated based on previous testing, while the Privacy Guide, Privacy Settings and Access History will be further developed. The team will also create the **User Flow and Information Architecture**, defining entry points, navigation, information hierarchy and connections between privacy information and relevant actions.

Main tasks:

- Refine the four prototype components.
- Develop the User Flow and Information Architecture.
- Define navigation and connections between the four stages.
- Review terminology and information consistency.

Expected outcome: Refined prototype screens, User Flow and Information Architecture.

6.2 Week 11 — Prototype Integration and Internal Pilot

Week 11 will integrate the four components into one interactive low- to medium-fidelity prototype. The team will review navigation, interaction and information consistency to ensure that the four stages work as one connected experience. An **internal pilot test** will then identify unclear tasks, missing information and major usability or technical problems. A shared testing protocol and data-recording template will be finalised to support consistent testing across team members.

Main tasks:

- Integrate the four components.
- Check navigation and interaction consistency.
- Conduct an internal pilot and address major problems.
- Finalise testing tasks, protocol and data-recording template.

Expected outcome: A test-ready integrated prototype and consistent testing protocol.

6.3 Week 12 — Final User Testing, Analysis and Iteration

In Week 12, the integrated prototype will be evaluated with approximately **eight participants**, with each team member conducting two sessions using the same protocol. Testing will evaluate the complete **SEE** → **UNDERSTAND** → **CONTROL** → **REVIEW**

journey, including whether users can find and understand privacy information, locate relevant controls, interpret Access History and move between the four stages. Results will be combined and reviewed by the team to identify repeated problems and prioritise final changes. The iteration will follow: **Testing Evidence → Identified Problem → Design Change**

Main tasks:

- Conduct eight user-testing sessions.
- Record task performance, comments and areas of confusion.
- Identify repeated patterns and key findings.
- Prioritise evidence-based changes and refine the prototype.

Expected outcome: Final testing findings and an evidence-based prototype iteration.

6.4 Week 13 — Final Documentation, Exhibit and Presentation

Week 13 will focus on completing the project deliverables. The team will finalise the **Design and Requirements Documentation**, explaining the relationship between research findings, design requirements, conceptual model, prototype development, evaluation and final design. The team will also prepare the **Final Exhibit and Presentation**, including the prototype demonstration, supporting materials, speaking roles and rehearsal. Report sections will be divided between members, with one member responsible for final integration and consistency.

Main tasks:

- Complete and review the final documentation.
- Finalise the low- to medium-fidelity prototype.
- Prepare and test the Final Exhibit.
- Prepare the Final Presentation and conduct a rehearsal.

Expected outcome: Final documentation, refined prototype, Final Exhibit and Final Presentation.

6.5 Team Responsibilities

The remaining work will combine individual responsibilities with shared team tasks. Each member will take the lead on one main area, while user testing, final iteration, documentation, exhibit preparation and presentation will be completed collaboratively.

Role	Main Responsibilities	Expected Output
Xinyue — Prototype Development	Refine the Privacy Guide, Privacy Settings and Access History; apply the final changes identified for the Privacy Label; prepare the main prototype screens.	Refined prototype screens

Yining — User Flow & Information Architecture	Develop the complete user flow; define entry points, navigation and information hierarchy; ensure that SEE → UNDERSTAND → CONTROL → REVIEW forms a clear journey.	User Flow + Information Architecture
Yuxuan — Prototype Integration	Connect the prototype screens into one interactive experience; maintain consistent navigation and interactions; prepare the prototype for the internal pilot and final testing.	Integrated low-/medium-fidelity prototype
Binyan — Testing Findings & Report Integration	Collect and organise testing data from all members; synthesise repeated problems and key findings; combine and review the final report for consistency.	Testing findings + integrated final report
All Members — User Testing	Each member conducts **2 user tests**, using the same testing protocol and recording structure.	Approx. **8 user-testing sessions**
All Members — Final Iteration	Review the testing findings together, decide which problems should be prioritised, and make evidence-based final design changes.	Final prototype iteration
All Members — Documentation	Divide and write sections of the Final Design and Requirements Documentation.	Final documentation
All Members — Final Exhibit & Presentation	Prepare exhibit materials, prototype demonstration, presentation slides, speaking roles, Q&A and rehearsal.	Final Exhibit + Final Presentation

6.6 Timeline Summary

Week	Focus	Key Activities	Main Output
Week 10	**Design**	Refine prototype, User Flow, Information Architecture	Refined screens + User Flow + IA
Week 11	**Integrate**	Prototype integration, navigation, internal pilot, testing preparation	Test-ready integrated prototype
Week 12	**Evaluate**	8 user tests, data analysis, findings, final iteration	Testing findings + refined prototype

Week 13	**Deliver**	Final documentation, exhibit, presentation, rehearsal	Final project deliverables

6.7 Project Risks and Management

Several risks may affect the remaining development. First, the four components may still feel like separate features after integration. To reduce this risk, the team will develop a shared User Flow and Information Architecture before completing the integrated prototype. Second, different team members will conduct user testing, which may create inconsistent data. A shared testing protocol and data-recording template will therefore be prepared before formal testing. Third, adding more privacy information may make the prototype too complex. The project will continue using progressive disclosure so that important information is visible first while additional explanation remains available when needed. Finally, some manufacturer privacy practices may not be clearly available or verifiable. The prototype will avoid making assumptions and will clearly communicate information that is unknown or not publicly specified.

By the end of Week 13, the team aims to deliver a tested and integrated low- to medium-fidelity prototype that demonstrates how home CCTV privacy information can become more **visible, understandable and actionable**, together with the final documentation, exhibit and presentation.

7. Team Reflection

7.1 Progress and Changes

At the beginning of the project, our problem space was relatively broad and covered different privacy issues around home CCTV systems. As the project developed, we realised that this scope was too large for one semester. We therefore narrowed our focus to the privacy relationship between CCTV users and manufacturers. We also found that our initial interviews did not provide enough diversity. To improve our research, we conducted **seven additional interviews** with participants from different backgrounds and experiences. The new data helped us compare different perspectives and update our key findings. These updated findings then provided a clearer direction for our design development. After updating the findings, we started developing prototypes. At this stage, we had four related design ideas, but they had not yet been fully connected. We therefore began by developing and testing the **Privacy Label** as the first prototype. While preparing the report, we realised that presenting four separate concepts did not clearly communicate our overall design direction. We needed one complete concept that could respond to the different findings together. As a result, we integrated the four ideas into one privacy journey: **SEE → UNDERSTAND → CONTROL → REVIEW**. This process changed our original plan, but it also gave the project a clearer structure. Our next step is to update the remaining prototype components and connect them through a shared user flow, so that they can be evaluated as one complete experience.

7.2 Team Collaboration

Our team used a combination of individual responsibilities and shared tasks. The responsibilities changed across different stages of the project based on the work required.

Stage	Team Member(s)	Contribution
Additional Interviews	Xinyue, Binyan, Yuxuan	Conducted the additional interviews to improve participant diversity.
Research Analysis	Yining	Organised interview data, updated the six findings, and added supporting literature.
Prototype Development	Yining	Developed the A/B/C Privacy Label prototypes and framework, and selected the first CCTV product.
Prototype Review	Binyan	Reviewed the A/B/C prototype content and testing plan.
Round 1 Testing	Yining, Xinyue, Binyan	Conducted the A/B/C prototype testing.
Round 2 Preparation	Yuxuan, Binyan, Xinyue	Yuxuan developed the multi-CCTV prototypes; Binyan supported product selection; Xinyue prepared the testing plan.
Round 2 Testing	Yining, Yuxuan, Xinyue	Conducted the multi-CCTV testing sessions.
Testing Analysis	Yuxuan	Organised findings from both rounds of prototype testing.
Report	Yining, Yuxuan, Binyan	Organised, reviewed and completed the interim report.

Presentation	Xinyue	Prepared the presentation slides.

Dividing responsibilities allowed us to work on different tasks at the same time. However, we also found that separate work could make different parts of the project less connected. This showed us the importance of regular team review and earlier integration of our design decisions.

References

Abdi, N., Ramokapane, K. M., & Such, J. M. (2019). *More than Smart Speakers: Security and Privacy Perceptions of Smart Home Personal Assistants*. 451–466. https://www.usenix.org/conference/soups2019/presentation/abdi?utm_source=chatgpt.com

Caven, P., Zhang, Z., Abbott, J., Ma, X., & Camp, L. (2024). Comparing the Use and Usefulness of Four IoT Security Labels. *Proceedings of the CHI Conference on Human Factors in Computing Systems*, 1–31. https://doi.org/10.1145/3613904.3642951

Chen, C. C., Shu, D., Ravishankar, H., Li, X., Agarwal, Y., & Cranor, L. F. (2024). Is a Trustmark and QR Code Enough? The Effect of IoT Security and Privacy Label Information Complexity on Consumer Comprehension and Behavior. *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems, CHI '24*, 1–32. https://doi.org/10.1145/3613904.3642011

Emami-Naeini, P., Agarwal, Y., Cranor, L. F., & Hibshi, H. (2020). Ask the Experts: What Should Be on an IoT Privacy and Security Label? *arXiv.Org*. https://www.proquest.com/docview/2354569785?pq-origsite=primo&searchKeywords=Ask%20the%20Experts%3A%20What%20Should%20Be%20on%20an%20IoT%20Privacy%20and%20Security%20Label%

3F%202020%20IEEE%20Symposium%20on%20Security%20and%20Privacy%2C%20447%E2%80%93464.&sourcetype=Working%20Papers

Emami-Naeini, P., Dixon, H., Agarwal, Y., & Cranor, L. F. (2019). Exploring How Privacy and Security Factor into IoT Device Purchase Behavior. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, CHI '19*, 1–12. https://doi.org/10.1145/3290605.3300764

Farke, F. M., Balash, D. G., Golla, M., Dürmuth, M., & Aviv, A. J. (2021). Are Privacy Dashboards Good for End Users? Evaluating User Perceptions and Reactions to Google's My Activity (Extended Version). *arXiv.Org*. https://www.proquest.com/docview/2535628918?pq-origsite=primo&searchKeywords=Are%20privacy%20dashboards%20good%20for%20end%20users%3F%20Evaluating%20user%20perceptions%20and%20reactions%20to%20Google%E2%80%99s%20My%20Activity.%20In%20Proceedings%20of%20the%2030th%20USENIX%20Security%20Symposium%20(pp.%20483%E2%80%93500).&sourcetype=Working%20Papers

Feng, Y., Yao, Y., & Sadeh, N. (2021). A Design Space for Privacy Choices: Towards Meaningful Privacy Control in the Internet of Things. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems, CHI '21*, 1–16. https://doi.org/10.1145/3411764.3445148

Samavi, R., & Consens, M. P. (2018). Publishing privacy logs to facilitate transparency and accountability. *Journal of Web Semantics*, *50*, 1–20. https://doi.org/10.1016/j.websem.2018.02.001

Schaub, F., Balebako, R., Durity, A. L., & Cranor, L. F. (2015). *A Design Space for Effective Privacy Notices*. 1–17.

https://www.usenix.org/conference/soups2015/proceedings/presentation/schaub?utm_source=chatgpt.com

Urquhart, L., Lodge, T., & Crabtree, A. (2019). Demonstrably doing accountability in the Internet of Things. *International Journal of Law and Information Technology*, *27*(1), 1–27. https://doi.org/10.1093/ijlit/eay015

Wang, Z., Yan, Y., Yan, Y., Chen, H., & Yang, Z. (2022). *CamShield: Securing Smart Cameras through Physical Replication and Isolation*. 3467–3484. https://www.usenix.org/conference/usenixsecurity22/presentation/wang-zhiwei?utm_source=chatgpt.com

Windl, M., Schlegel, M., & Mayer, S. (2024). Exploring Users' Mental Models and Privacy Concerns During Interconnected Interactions. *Proceedings of the ACM on Human-Computer Interaction*, *8*(MHCI), 1–23. https://doi.org/10.1145/3676504

Yao, Y., Basdeo, J. R., McDonough, O. R., & Wang, Y. (2019). Privacy perceptions and designs of bystanders in smart homes. *Proceedings of the ACM on Human-Computer Interaction*, *3*(CSCW). https://doi.org/10.1145/3359161

Zheng, S., Apthorpe, N., Chetty, M., & Feamster, N. (2018). User Perceptions of Smart Home IoT Privacy. *Proceedings of the ACM on Human-Computer Interaction*. https://dl.acm.org/doi/10.1145/3274469

AI Use Statement

Generative AI tools, primarily ChatGPT, were used in a limited and supporting capacity throughout this project. The main purpose of AI use was to support language refinement, improve the clarity of written communication, and assist with the organisation and presentation of existing project content.

AI support was primarily used to:

- improve grammatical accuracy and correct language errors;
- refine sentence structure and improve readability;

- translate or refine wording between Chinese and English;
- refine sentence structure and improve the clarity of report sections;
- organise existing project content into clearer headings, tables, and report structures;
- improve the wording and structure of presentation materials;
- review whether written explanations were clear and logically connected.

AI was not used to generate original arguments, research findings, or core content.

The final report reflects the authors' own understanding, with AI serving only as a language support tool.

Appendices

Appendix A — Interview Questions

A semi-structured interview was conducted to explore participants' experiences with home CCTV systems, their understanding of CCTV data practices, and their expectations regarding privacy, transparency, and control. The questions below were used as an interview guide. Follow-up questions were asked when relevant based on participants' responses.

A.1 CCTV Usage：

Main Question: Have you ever used a home CCTV or smart security camera? Can you tell me a little about how you use it?

Follow-up Questions

- What do you mainly use it for?
- How often do you check the camera or recordings?
- Who else in your home uses the CCTV?

A.2 Data Storage:

Main Question: Where do you think your CCTV videos are stored?

Follow-up Questions

- How do you know that?
- Do you know if your CCTV uses cloud storage or local storage?
- Do you know how long the recordings are kept?
- Have you ever changed any storage settings?

A.3 Data Access:

Main Question: Who do you think can watch your CCTV recordings?

Follow-up Questions

- Do you think anyone else can see them besides you?
- Why do you think that?
- What about the CCTV company? Do you think the company or its employees can access your recordings?
- In what situations do you think they should be allowed to access your recordings?
- Do you think they should ask for your permission first?
- If the company accessed your recordings, would you want to know? Why?

A.4 Privacy Information:

Main Question: When you set up your CCTV, did you read any information about privacy?

If yes:

- Was the information easy to understand?
- What do you remember from it?
- Was there anything that was difficult to understand?

If no:

- Why did you not read it?
- Was it too long, too difficult, or hard to find?
- What would make you more likely to read privacy information?

A.5 Trust:

Main Question: What would make you lose trust in a CCTV company?

Follow-up Questions

- Would a data leak change how you use the camera? Why?
- What if the company accessed your recordings without your permission?
- What would the company need to do to keep your trust?

A.6 Transparency Needs:

Main Question: Is there anything you would like to know about where your CCTV videos go or who can see them?

Follow-up Questions

- What information would you most like to know?
- Why is that information important to you?
- When would you like to see this information — before buying the camera, during setup, or while using it?
- How would you like this information to be shown to you?

- Would you prefer short information, pictures, labels, or more detailed information?

A.7 Mental Model Activity:

Participants were asked to complete a short drawing activity to explore their understanding of CCTV data flow.

Instruction

Imagine you are outside your home and you open the CCTV app on your phone to watch your camera. Can you quickly draw how you think the video gets from the camera to your phone? There is no right or wrong answer. Just draw what you think happens.

Follow-up Questions

- Can you explain your drawing to me?
- What do you think happens between the camera and your phone?
- Do you think the video goes through any other system or company?
- Is there any part of this process that you are not sure about?

Appendix B — Participant Background

Participant	CCTV Experience	Context	Key Characteristic
P1	Current user	Home	Uses CCTV for pet/family monitoring
P2	Current user	Home	Limited understanding of data flow
P3	Potential user	Lives alone	Security need but concerned about privacy
P4	Current user	Family home	Parent using CCTV for family monitoring
P5	Current user	Home	Regular CCTV use
P6	Current user	Home	Some awareness of cloud/data flow
P7	Current/previous user	Home	CCTV user perspective
P8	Current user	Home	Some technical awareness of data flow
P9	Current/previous user	Home	Privacy-information perspective

P10	Current/previous user	Home	Privacy and data-practice concerns
P11	CCTV installer	Professional	Technical/stakeholder perspective
P12	Current user	Home	Everyday CCTV user perspective

Appendix C — Interview Selected Quotes

The following selected quotes are taken directly from the interview data. They provide supporting evidence for the six key findings identified through the user research.

Participant	Selected Quote	Related Finding
P1	“I'm not sure whether they technically can, but I definitely wouldn't want them to access my recordings without my permission.”	F2 — Manufacturer & External Access
P1	“If the company stole or accessed my recordings without my permission... I would completely lose trust in them.”	F3 — Privacy Risks & Trust
P2	“I know that I can open the app on my phone and watch the camera, but I don't really know how the video actually gets from the camera to the app.”	F1 — Storage & Data Flow
P2	“If the company needs to access a customer's recordings, they should let the customer know and get permission first.”	F6 — User Control
P3	“The most important thing for me is probably who can see the video. I also want to know where it's stored.”	F5 — Information Needs

P3	“But I probably wouldn't read the entire privacy policy. They're usually really long and difficult to understand.”	F4 — Information Complexity
P4	“I'm actually not sure whether they technically can. I just don't think they should be allowed to access private recordings from inside our home.”	F2 — Manufacturer & External Access
P4	“If the company accessed or took my recordings without my permission, I would lose trust in them.”	F3 — Privacy Risks & Trust
P5	“Then the video probably goes through the manufacturer's system or server. But I'm not completely sure what happens in the middle.”	F1 — Storage & Data Flow
P5	“I don't want a very long privacy document.”	F4 — Information Complexity
P6	“Honestly, I’m not completely sure. I think some of them are stored in the cloud because I can watch the camera when I’m outside. But there’s also a memory card inside it, so maybe the recordings are stored in both places.”	F1 — Storage & Data Flow
P6	“If someone new accesses the camera, I’d want a notification.”	F6 — User Control
P7	“I assume the company can access them in some situations as well. What I don’t know is whether their employees can just open the videos or whether they need special permission.”	F2 — Manufacturer & External Access
P7	“I’d prefer a short privacy summary when setting up the camera. Something visual and easy to scan.”	F5 — Information Needs

P8	“Possibly a cloud storage provider or technical support staff. That’s something I hadn’t really thought about before. There could be another company involved that I don’t even know about.”	F2 — Manufacturer & External Access
P8	“Give me more control over what gets uploaded.”	F6 — User Control
P9	“I think they are probably stored in the cloud, but maybe some of them are stored on a memory card. I am not completely sure.”	F1 — Storage & Data Flow
P9	“I would prefer a simple page in the app and a label that clearly shows where the videos are stored, how long they are kept, who can access them, and how to delete them.”	F5 — Information Needs
P10	“Actually, I am not sure, but I would prefer that the company could not see my recordings.”	F2 — Manufacturer & External Access
P10	“If the company wants to access my recordings, it must let me know.”	F6 — User Control
P11 — Installer	“If we need to view a customer’s footage or information, we must get their consent first. We cannot simply watch it whenever we want.”	F2 — Manufacturer & External Access
P11 — Installer	“Users should remain in control of their recordings, and we should ask for consent before viewing them.”	F6 — User Control

P12	“I do not know whether people working for the CCTV company can view the videos, or under what circumstances they would be allowed to do that. I would like the company to explain this clearly.”	F2 — Manufacturer & External Access
P12	“I do not want to read a long document just to understand the basic privacy information.”	F4 — Information Complexity

Appendix D — Affinity Mapping

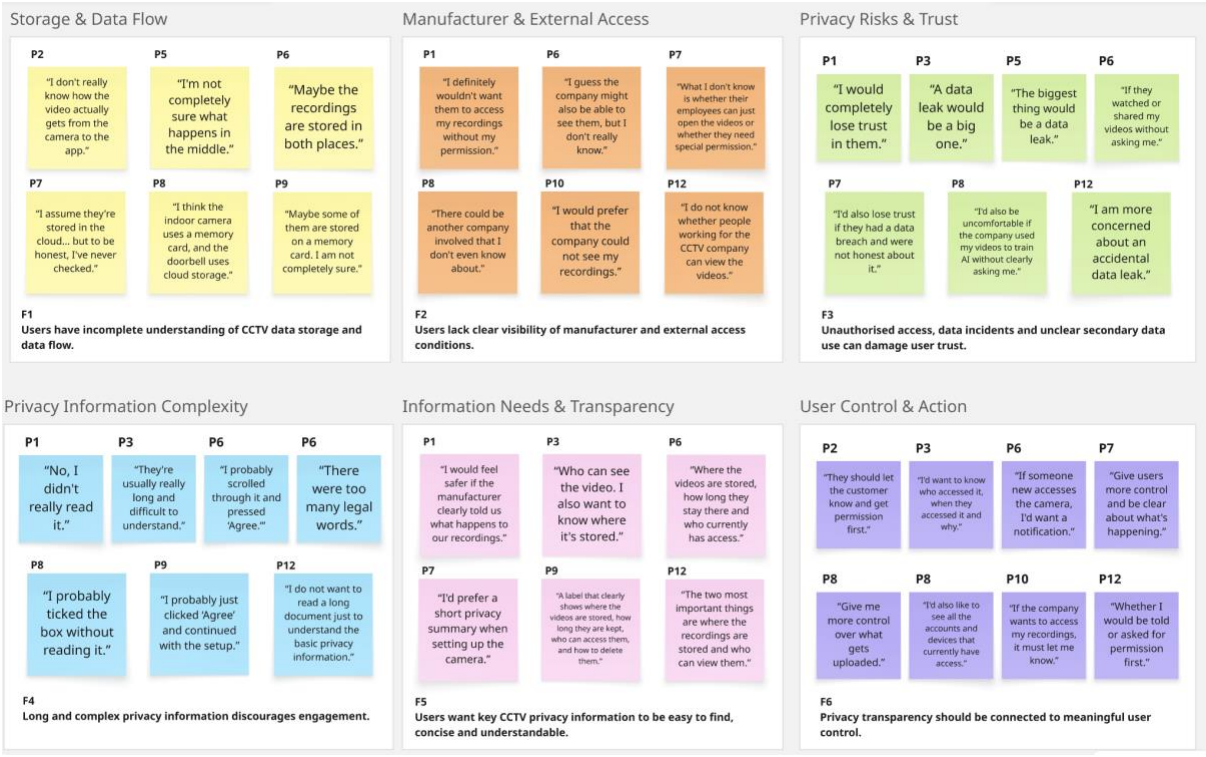


Figure 5. Affinity mapping of interview data used to identify repeated themes and develop the six key research findings.

Appendix E — Finding–Evidence Matrix

Finding	Supporting Participants	Key Evidence	Design Implication

F1 — Incomplete understanding of CCTV data storage and data flow	P1, P2, P3, P5, P6, P7, P8, P9, P10, P11, P12	Participants showed uncertainty about cloud/local storage, retention periods, and how recordings move from the camera to the app.	Clearly explain storage location, retention and data flow.
F2 — Limited visibility of manufacturer and external access	P1, P2, P3, P4, P6, P7, P8, P9, P10, P11, P12	Participants were unsure whether manufacturers, employees, support staff or cloud providers could access recordings and under what conditions.	Show manufacturer access, third-party access and permission conditions.
F3 — Privacy risks can damage user trust	P1, P2, P3, P4, P5, P6, P7, P8, P9, P10, P11, P12	Unauthorised access, data leaks, undisclosed sharing and unclear secondary data use were repeatedly linked to reduced trust.	Make access, sharing, consent and privacy risks more visible.
F4 — Long and complex privacy information discourages engagement	P1, P2, P3, P4, P6, P7, P8, P9, P10, P11, P12	Many participants skipped or only partly read privacy information because it was long, complex or difficult to understand.	Use concise information, plain language and progressive disclosure.
F5 — Key privacy information should be easy to find and understand	P1–P12	Participants prioritised information such as storage, access, retention and permissions, and many preferred short summaries or simple visual information.	Provide a concise Privacy Label with access to more detail.
F6 — Transparency should connect to meaningful user control	P2, P3, P6, P7, P8, P9, P10, P11, P12	Participants wanted permission controls, access notifications, deletion options, upload controls and access history.	Connect privacy information with Privacy Settings and Access History.

Appendix F — Prototype A/B/C

PROTOTYPE A

RING PAN-TILT INDOOR CAMERA

FACTS ONLY - No privacy score or rating

PRIVACY INFORMATION

Who Can Access?	You and the Shared Users you authorise can access the camera.
Video Storage	Recorded videos are stored in the Ring cloud when recording features are used with an eligible Ring subscription.
Manufacturer Access	Optional End-to-End Encryption (E2EE) can be enabled. When E2EE is enabled, Ring services cannot access your encrypted video content.
Data Sharing	Some personal information may be processed by Ring affiliates and service providers for defined purposes such as storage, customer service and analytics.
Video Retention	For event videos, you can choose a storage period from **1 to 180 days.** The default is **30 days**.
Your Controls	Manage Shared Users; delete recordings; change event-video retention time; enable optional E2EE.
Physical Privacy	Closing the physical Privacy Cover turns off the camera and microphone.

Research question: Can users understand the privacy information from facts alone?

Figure 6. Prototype A — Facts Only used in the Round 1 Privacy Label comparison.



Figure 7. Prototype B — Rating Label used in the Round 1 Privacy Label comparison.

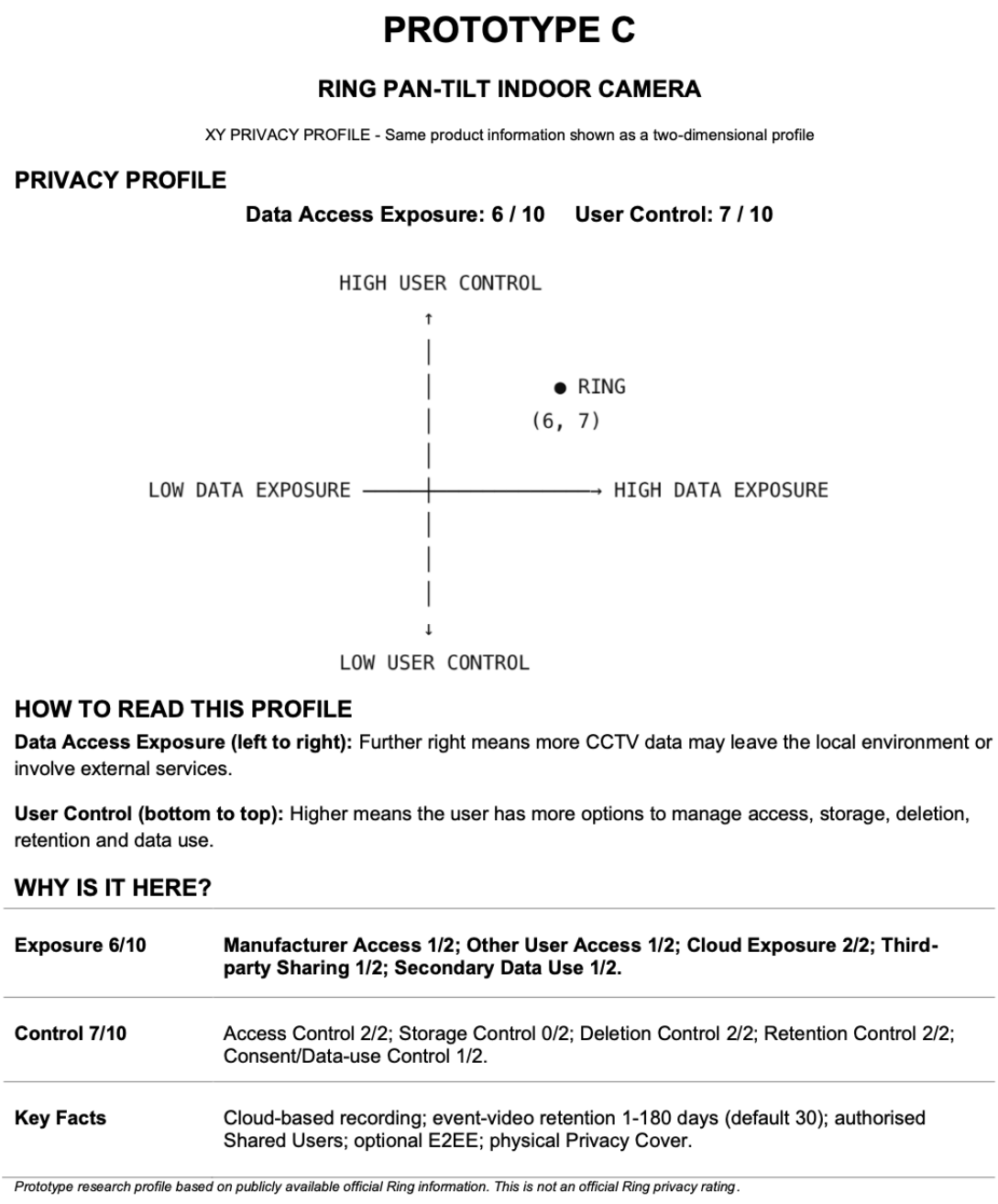


Figure 8. Prototype C — Visual Profile used in the Round 1 Privacy Label comparison.

Appendix G — Round 1 - Testing Raw Data

Participant	Presentation Order	Accuracy A / B / C	Time A / B / C	Confidence A / B / C	Preference Rank A / B / C

R1-P01	B → C → A	2/3 / 3/3 / 3/3	62s / 50s / 42s	3/5 / 4/5 / 4/5	1 / 3 / 2
R1-P02	A → B; C also evaluated	2/3 / 2/3 / 2/3	2m / 2m / 1m30s	3/5 / 3/5 / 2/5	2 / 1 / 3
R1-P03	A → B → C	3/3 / 3/3 / 3/3	42s / 55s / 67s	5/5 / 4/5 / 3/5	1 / 2 / 3
R1-P04	B → C → A	2.5/3 / 3/3 / 3/3	58s / 30s / 39s	4/5 / 5/5 / 4/5	3 / 1 / 2
R1-P05	B → A → C	3/3 / 3/3 / 2/3	82s / 72s / 103s	4/5 / 5/5 / 3/5	2 / 1 / 3
R1-P06	C → B → A	3/3 / 3/3 / 2/3	69s / 58s / 111s	4/5 / 5/5 / 3/5	3 / 1 / 2

Appendix H — Muti_Level_Privacy_Label

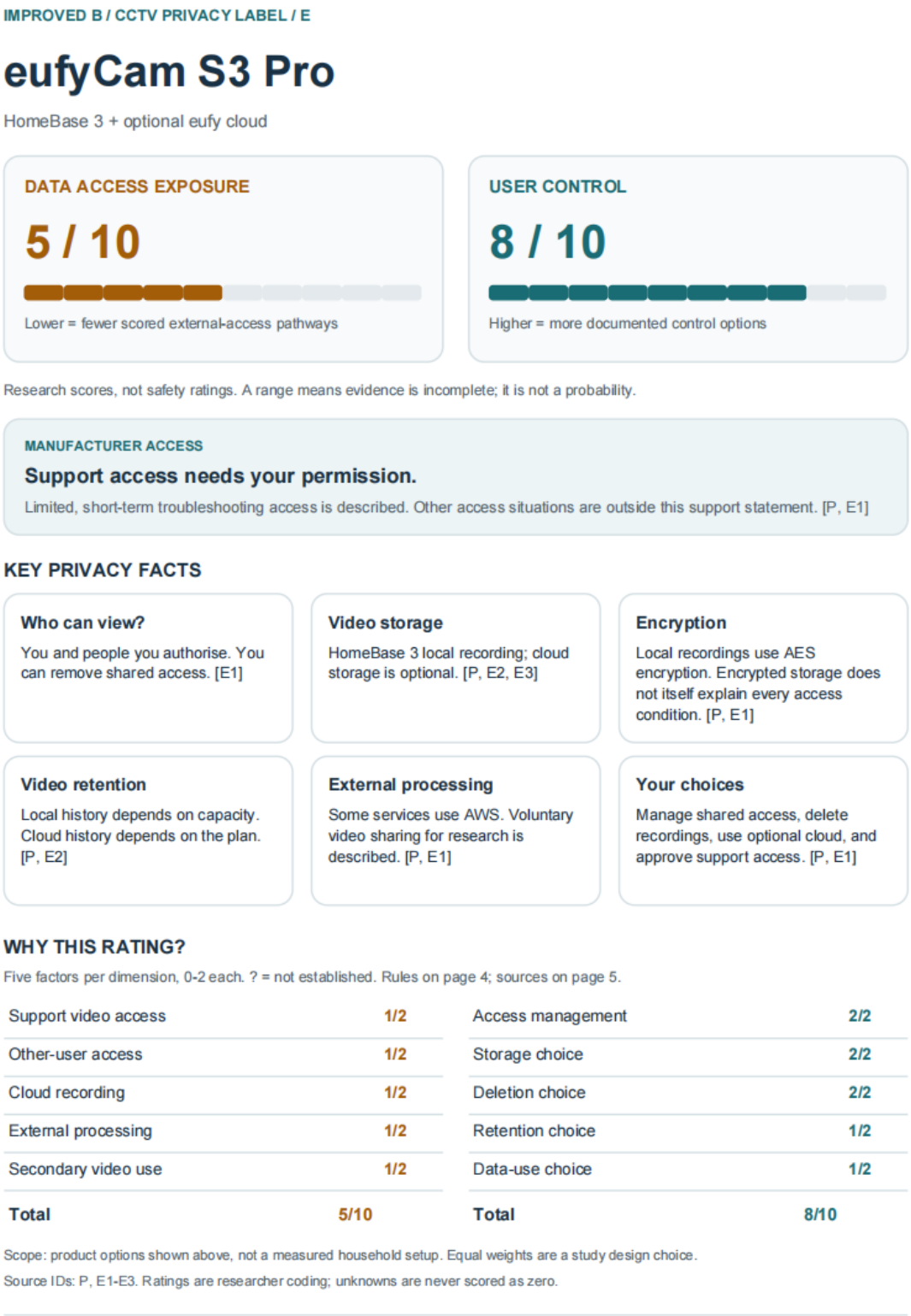


Figure 9. Privacy Label prototype for eufyCam S3 Pro.

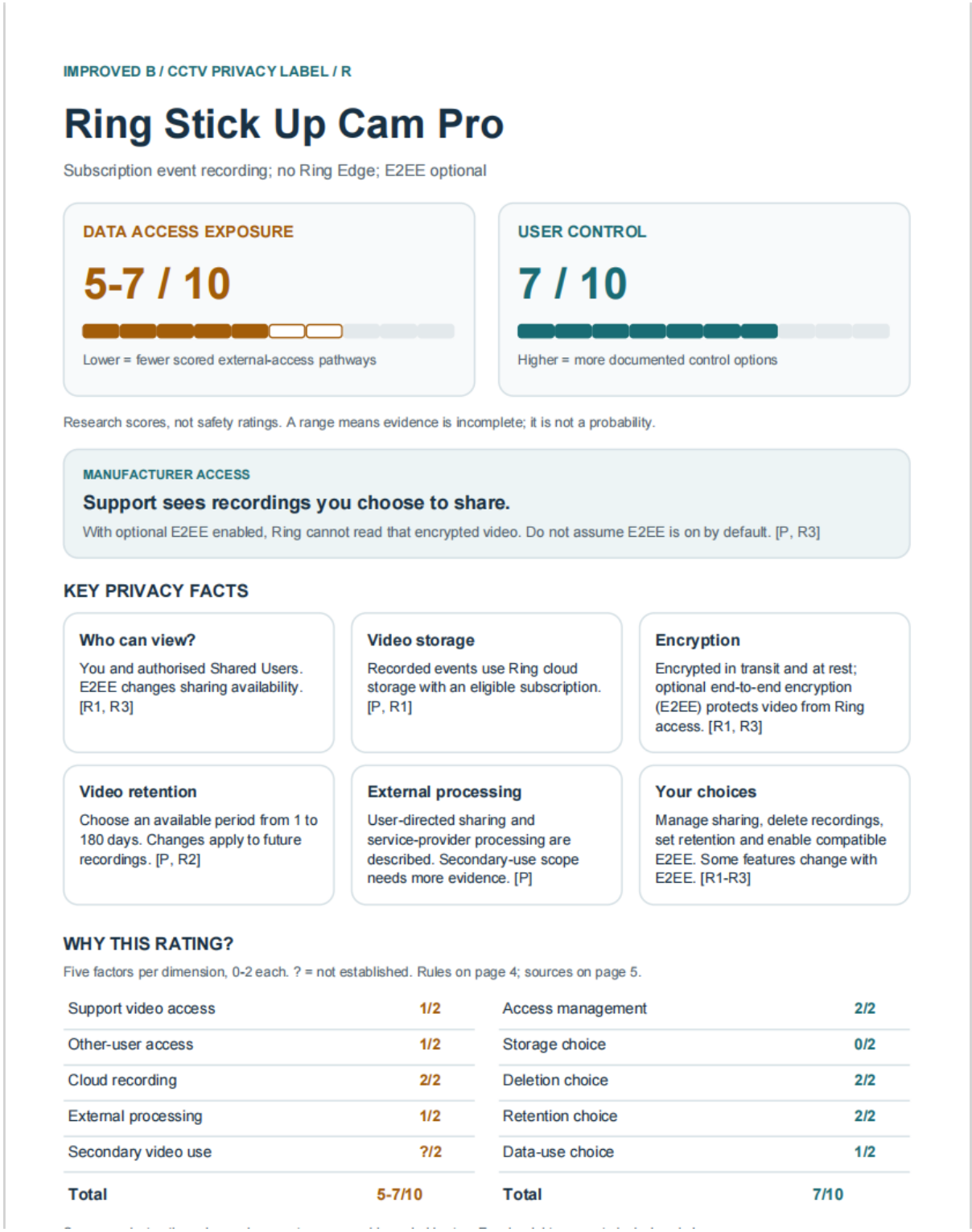
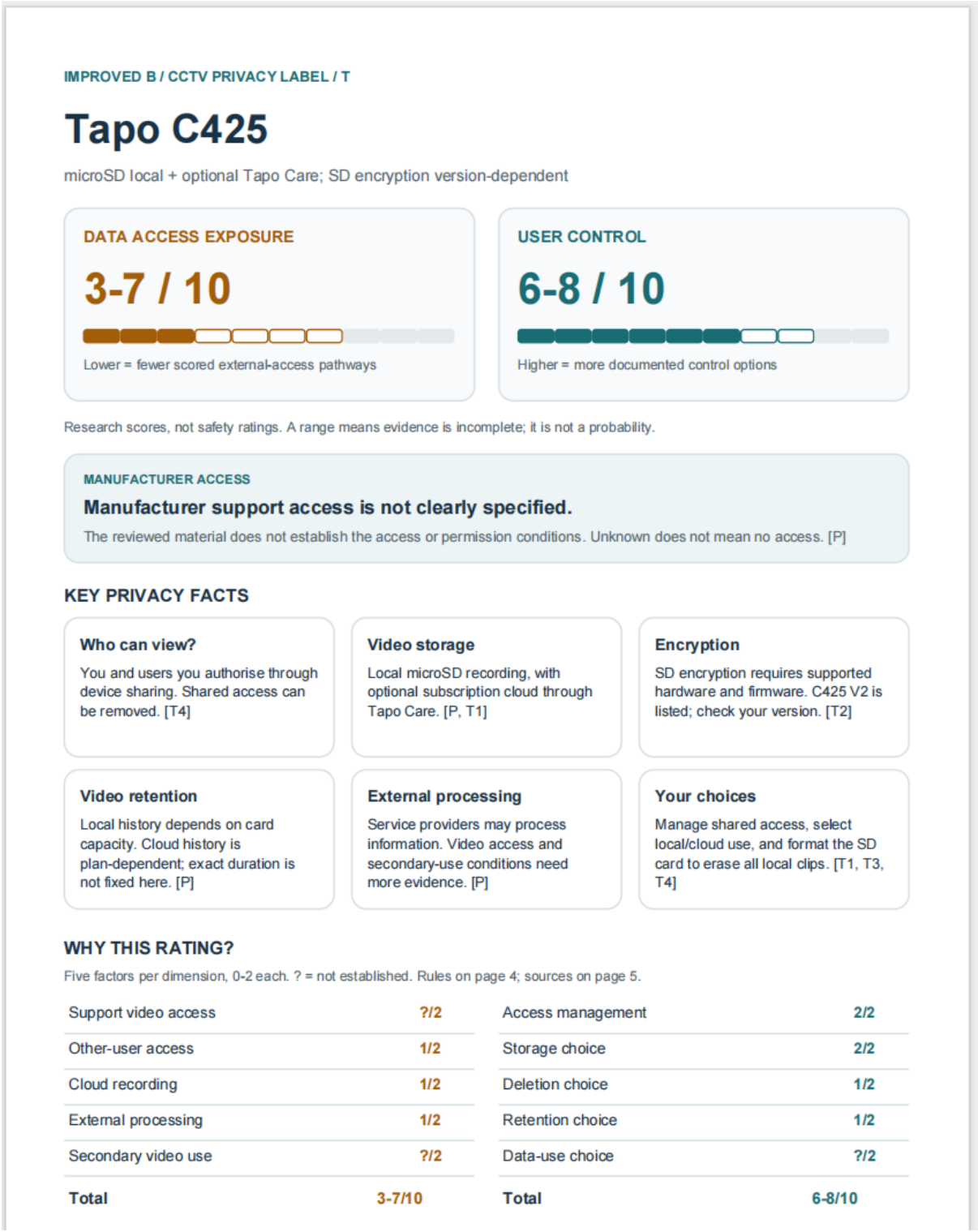


Figure 10. Privacy Label prototype for Ring Stick Up Cam Pro.



IMPROVED B / CCTV PRIVACY LABEL / T

Tapo C425

microSD local + optional Tapo Care; SD encryption version-dependent

DATA ACCESS EXPOSURE

3-7 / 10

Lower = fewer scored external-access pathways

USER CONTROL

6-8 / 10

Higher = more documented control options

Research scores, not safety ratings. A range means evidence is incomplete; it is not a probability.

MANUFACTURER ACCESS

Manufacturer support access is not clearly specified.

The reviewed material does not establish the access or permission conditions. Unknown does not mean no access. [P]

KEY PRIVACY FACTS

Who can view?
You and users you authorise through device sharing. Shared access can be removed. [T4]

Video storage
Local microSD recording, with optional subscription cloud through Tapo Care. [P, T1]

Encryption
SD encryption requires supported hardware and firmware. C425 V2 is listed; check your version. [T2]

Video retention
Local history depends on card capacity. Cloud history is plan-dependent; exact duration is not fixed here. [P]

External processing
Service providers may process information. Video access and secondary-use conditions need more evidence. [P]

Your choices
Manage shared access, select local/cloud use, and format the SD card to erase all local clips. [T1, T3, T4]

WHY THIS RATING?

Five factors per dimension, 0-2 each. ? = not established. Rules on page 4; sources on page 5.

Support video access	?/2	Access management	2/2
Other-user access	1/2	Storage choice	2/2
Cloud recording	1/2	Deletion choice	1/2
External processing	1/2	Retention choice	1/2
Secondary video use	?/2	Data-use choice	?/2
Total	**3-7/10**	**Total**	**6-8/10**

Figure 11. Privacy Label prototype for Tapo C425.

Appendix I — Round 2 - Testing Raw Data

Participant	Product Order	Comparison Accuracy	Comparison Time	Initial → Final Choice	Choice Confidence Before → After	Final System Confidence / Clarity	Reported Trust After Evidence
R2-P01	eufy → Ring → Tapo	4/4	3m 06s	eufy → eufy	4/5 → 4/5	5/5	Higher
R2-P02	Ring → Tapo → eufy	3/4	5m 18s	Ring → eufy	3/5 → 4/5	4/5	Higher
R2-P03	Ring → Tapo → eufy	4/4	2m 18s	eufy → eufy	3/5 → 4/5	4/5	Higher
R2-P04	eufy → Ring → Tapo	4/4	2m 20s	eufy → eufy	3/5 → 4/5	4/5	Higher
R2-P05	Ring → Tapo → eufy	3/4	2m 45s	Tapo → Tapo	3/5 → 3/5	4/5	Same
R2-P06	eufy → Ring → Tapo	4/4	2m 41s	eufy → eufy	4/5 → 5/5	4/5	Higher